

Aaron Detre

(510) 421-4577 | aarondetre@gmail.com | [Portfolio](#) | linkedin.com/in/aarondetre

Education

Northeastern University, Khoury College of Computer Sciences | Boston, MA
B.S. Computer Science, Minor in Psychology, summa cum laude | April 2026 | **GPA: 3.91**

Skills

Languages: Java, TypeScript, JavaScript, SQL, HTML, CSS
Frameworks: React, Next.js, Angular, Spring Boot, Node.js, Express.js
Tools & Platforms: Git, GitHub Actions, Docker, AWS CodePipeline, MongoDB
Operating Systems: Linux (Ubuntu), Windows
Engineering Practices: Agile, CI/CD, Pair Programming, Unit Testing

Experience

Full-Stack Software Developer Co-op (Angular, Typescript, Java Spring Boot)

Technology Enhanced Learning in Science, UC Berkeley Jan. 2025 - June 2025

- Developed production features in an existing Angular codebase for WISE, a globally used, open-source virtual learning platform supporting K–12 science education and ongoing research.
- Collaborated directly with education researchers and classroom teachers to translate research goals and instructional needs into software solutions used in real classrooms.
- Designed and implemented guest-accessible units with registration-free access, allowing WISE to function as a survey tool for studies requiring more flexible, activity-based data collection.
- Built teacher-facing summary visualizations that aggregate class-wide student responses, including a summary display for an AI chat activity highlighting the most and least common ideas detected, making important but underrepresented ideas more visible to instructors and researchers.
- Improved developer workflows by automating internationalization annotation generation using GitHub Actions, reducing manual effort and preventing recurring errors in pull requests.
- Wrote unit tests, participated in peer code reviews, and iterated based on feedback from engineers.

Projects

Recipe Social Network (React, Next.js, Node.js, MongoDB)

Nov. 2025 - Dec. 2025

- Designed and built a full-stack recipe-sharing platform with authentication, allowing users to create, review, and remix recipes while following others for a personalized feed.
- Implemented scalable ingredient handling in response to user feedback, supporting numeric and non-numeric quantities and dynamic scaling via user-defined multipliers.
- Developed non-relational MongoDB schemas and implemented Express.js REST endpoints to support search functionality across both internal data and an external recipe API.

Programming Language Implementation (Java, JUnit)

Sept. 2025 - Dec. 2025

- Designed and implemented a 12,000-line object-oriented Java system for a module-based programming language, including an abstract syntax tree, parser, semantic validity checker, and CESK machine.
- Implemented support for programs combining typed and untyped modules with cross-module imports, enforcing soundness via runtime proxies and ensuring safe interactions at module boundaries.
- Applied the Strategy Pattern to improve maintainability by modularizing 25 CESK transition cases.
- Developed a comprehensive testing suite with 100+ integration tests written in the target language alongside JUnit tests, with automated execution via a Bash-based test runner.

Interests

Urban and Nature Photography | Independent Films | Bread Baking

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Education

Northeastern University, Khoury College of Computer Sciences | Boston, MA
B.S. Computer Science, Minor in Psychology | Expected May 2026 | **GPA: 3.9**

Skills

Languages: TypeScript, Java, JavaScript, SQL, HTML, CSS

Frameworks: React, Next.js, Angular, Spring Boot, Node.js, Express.js

Tools & Databases: Git, GitHub Actions, MongoDB, Docker

Development Practices: Pair Programming, Agile, CI/CD, RESTful API Design

Experience

Full-Stack Software Developer Co-op

Technology Enhanced Learning in Science, UC Berkeley

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- Designed and built guest-accessible units enabling users to access lessons without creating an account, implementing both frontend components and backend API endpoints.
- Implemented data visualization displays to provide insights into student progress and engagement.
- Enhanced the CI/CD pipeline by configuring GitHub Actions to automatically generate internationalization annotations on pull requests, reducing manual i18n overhead.
- Presented technical updates to education researchers at Berkeley and partner institutions.

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- Implemented search from both an external recipe API and a MongoDB database of user-generated content accessed with Express.js REST endpoints.
- Designed non-relational schema modeling relationships between users, recipes, reviews, and follows.

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Sept. 2025 - Dec. 2025

Course project taught by Prof. Matthias Felleisen (co-creator of Racket)

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- Implemented support for systems that combine typed and untyped modules with cross-module imports.
- Validated code before runtime, including variable scope verification, type checking, import validation.
- Enforced soundness at runtime using proxies to catch type violations at module boundaries.
- Built the CESK machine using Strategy pattern with 25 transition cases, improving code maintainability by delegating each case to separate strategy classes.
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USA resume ^^^

Canada resume vvv

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Development Practices: Pair Programming, Agile, CI/CD, RESTful API Design

****Additional:**** Authorized to work in Canada without sponsorship

Old stuff:

Aaron Detre

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Availability: May 2026

Education

Northeastern University, Boston, MA

Sept. 2021 - May 2026 (expected)

Khoury College of Computer Sciences

GPA: 3.88/4.0

Candidate for a Bachelor of Science in Computer Science, Minor in Psychology

Academic Honors: Dean's list

Related courses: Object-Oriented Design | Algorithms and Data | Programming Languages

Database Design | Network Fundamentals | Computer Systems

Computer Knowledge

Languages: TypeScript | JavaScript | Java | HTML | CSS | Python | SQL
Frameworks: Angular | React | Spring Boot
Software: VS Code | Ubuntu | GitHub | Docker | Balsamiq
Systems: Linux | Windows

Projects

[Photography Site](#)

June 2025 - Sept. 2025

- Built a responsive photography portfolio website using React and Typescript.
- Designed and implemented a clean, intuitive UI with CSS.
- Optimized site performance with image compression tools and lazy loading.
- Deployed the site to Vercel, enabling automated builds from GitHub pushes.

Web Crawler

Oct. 2024

- Created a Python web crawler that sends HTTP requests to log into and traverse a simulated social media site, parsing each page's HTML to locate hidden flags.
- Implemented robust crawling logic by tracking visited pages to prevent cycles, managing session cookies for authentication, and handling HTTP status codes.

Reversi Game

Oct. - Dec. 2023

- Developed a fully playable version of the board game Reversi in Java, applying the MVC model and leveraging design patterns including Observer, Decorator, and Strategy to ensure maintainability and scalability.
- Designed a visually appealing user interface using the Java Swing library, enhancing the overall user experience.
- Wrote comprehensive unit tests using JUnit to validate game logic and ensure reliability.

Work Experience

Full-Stack Web Developer Co-op

Technology Enhanced Learning in Science, UC Berkeley

Jan. - June 2025

- Served as a full-stack developer on a hybrid Agile team, working on an open-source virtual learning platform.
- Built front-end features with Angular + TypeScript and back-end services with Java Spring.
- Implemented new functionality, including guest-accessible classroom units, teacher-facing student data summaries, and timed learning activities.
- Enhanced the CI/CD pipeline by using GitHub Actions to automatically generate internationalization annotations on pull requests.
- Practiced pair programming to improve code quality, share knowledge, and accelerate feature development.

Print Version



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Aaron Detre [before co-op]

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Availability: January - August 2025

Education

Northeastern University, Boston, MA Sept. 2021 - May 2026 (expected)
Khoury College of Computer Sciences **GPA: 3.8/4.0**
Candidate for a Bachelor of Science in Computer Science
Academic Honors: Dean’s list

Related courses: Object-Oriented Design | Algorithms and Data | Computer Systems
Theory of Computation | Network Fundamentals | Database Design

The College Preparatory School, Oakland, CA

Sept. 2017 - May 2021

Related courses: Computer Science Foundations | Computer Programming with Python

GPA: 3.7/4.0

Computer Knowledge

Languages: Java | Python | C | SQL | HTML | CSS | C# | Racket
Software: Visual Studio Code | IntelliJ | MySQL | Github | Unity | Virtual Box
Systems: Windows | Linux

Projects

3D Video Game - [Itch.io Link](#) June - Aug. 2024

- Designed and published a multi-level 3D video game made in Unity, showcasing strong game design principles.
- Wrote efficient C# scripts to handle player input, sound effects, particle systems, and level progression, ensuring a polished gameplay experience.

Chat Server April 2024

- Developed a Linux terminal-based chat server in C, utilizing sockets and file descriptors to enable real-time communication between two clients over the internet.
- Implemented an automatic word filter that detects and replaces certain words before transmitting messages, enhancing my understanding of string manipulation in C.
- Utilized GDB to debug and optimize the code, improving the server's reliability and performance.

Reversi Game Oct. - Dec. 2023

- Developed a fully playable version of the board game Reversi in Java, writing over 2,000 lines of clean, readable, and well-documented code.
- Applied the MVC model and leveraged design patterns such as the Observer, Decorator, and Strategy patterns to structure the application for maintainability and scalability.
- Designed a visually appealing user interface using the Java Swing library, enhancing the overall user experience.

Work Experience

Visitor Experience Associate - PHI Studio Oct. 2022 - April 2023

- Greeted guests and ensured their safety throughout a virtual reality experience, demonstrating strong interpersonal skills.
- Collaborated effectively with team members, taking on additional responsibilities as needed to ensure smooth operations and a positive guest experience.

- Stuff to add from co-op

- Langs/tech
 - JS, TS, Java Spring, Angular, Docker, AWS(?)
 - Github (Github Actions), Git
- Refactored code (convert to standalone/refactor)
 - Big refactor on 1000 line summary display component
- Implemented new features
 - Multiple AI endpoints
 - Student-friendly text for ideas authoring
 - Initial glucose stored for leaf glucose simulation model
 - Changed glucose animation
 - Display different pics for glucose levels through URL
 - Summary displays
 - Dialog guidance
 - Survey units
 - Timed steps
- Implemented new features for the site such as units that don't require an account to access, visual displays summarizing student data for teachers to view, and activities with time limits.
- CI/CD stuff
 - I18n Internationalization generate messages automatically updated through Github Actions
 - Leveraged Github Actions to improve the CI/CD pipeline by automatically generating internationalization annotations on each pull request.
- Lo-fi mockups for new features
 - Match summary display
- Made art assets for plant simulations
- Publishing npm packages
- Pair programming
- Presented tech updates to researchers
- Attended research meetings and shared ideas
- Daily standups
- Remote work + slack + zoom
- In person work

Technical: Front-end development using Angular framework. Backend/API development with Spring framework, Redis, MySQL. Application deployment with Docker and AWS codepipeline. Basic server administration skills like installation and security using DigitalOcean. Learned more git commands and functions on GitHub to collaborate with other developers. Introduction to NLP and discussing AI-related tasks. Mockup initial designs on paper and on Balsamiq.

Soft-skills: communicated with developers, designers, machine learning scientists, researchers, teachers, students. Participated in agile methodologies like daily stand ups, retrospectives, pair/mob programming. Presented work to researchers at Berkeley and at other institutions. Observed classroom implementation of software and gathered feedback from users.

Strengths: Ability to reason and understand complex concepts. Listens to others well and asks great questions. Demonstrates curiosity and ability to accept feedback. Solid programming skills and willingness to keep improving.

Areas of improvement: Presentation skills could use more improvement. Sometimes Aaron was hard to hear and understand because he was soft-spoken and talked fast. Speak slower and louder and make sure that people are following along. Communicate more confidently about all the great work that you did.

Projects to add:

- Survey Units?
- Recipe Site
 - Nov-dec 2025
 - Designed a responsive, mobile-first, multi-page web application in React for recipe creating, sharing, and reviewing.
 - Implemented the frontend with Next.js, the backend with Node.js, and the database with MongoDB.
 - Styled the site using CSS and React Bootstrap to match my design mockups.
- AI Recipe Site entry:

Recipe Site (React, Next.js, Node.js, MongoDB) - November - December 2025

- Developed a full-stack recipe social network with user authentication, enabling users to create, remix, review, and share recipes with a community of followers
- Built a dynamic recipe scaling feature that automatically recalculates ingredient quantities based on user-specified serving multipliers, handling both scalable (numerical) and non-scalable ingredient formats
- Integrated third-party recipe API alongside user-generated content, implementing a unified search system that queries both external recipes and MongoDB database of user submissions, reviews, and remixed versions

Claude resume:

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Experience

Full-Stack Software Developer Co-op

Technology Enhanced Learning in Science, UC Berkeley | Jan. 2025 - June 2025

- Developed full-stack features for an open-source virtual learning platform using Angular, TypeScript, Java Spring Boot, MySQL, and Redis, deployed via Docker and AWS CodePipeline
 - Architected and built survey units feature enabling teachers to collect student feedback within lessons, implementing both frontend UI components and backend API endpoints with data persistence
 - Refactored 1,000+ line Angular component to improve maintainability and implemented multiple teacher-facing data visualization displays for classroom insights
 - Enhanced CI/CD pipeline by configuring GitHub Actions to automatically generate internationalization annotations on pull requests, reducing manual i18n overhead
 - Collaborated on 3-person development team using Agile practices including daily standups, pair programming, and sprint retrospectives; presented technical updates to education researchers at Berkeley and partner institutions
-

Projects

Recipe Social Network (React, Next.js, Node.js, MongoDB) | Nov. 2025 - Dec. 2025

- Built a full-stack recipe sharing platform with user authentication, enabling users to create, review, and remix recipes while following other users for a personalized feed
- Developed dynamic recipe scaling system that automatically recalculates ingredient quantities based on user-specified multipliers, supporting both scalable (numeric) and non-scalable ingredient formats
- Integrated third-party recipe API with Express.js REST endpoints alongside MongoDB database of user-generated content, implementing unified search across both external and internal recipe sources
- Designed relational data model handling users, recipes, reviews, follows, and recipe remixes with proper foreign key relationships

Programming Language Implementation (Java, JUnit) | Sept. 2025 - Dec. 2025

Course project taught by Dr. Matthias Felleisen (co-creator of Racket)

- Designed and implemented parser, type checker, and CESK abstract machine semantic model for a Scheme-based class and module system in Java (12,000 lines)
- Built mixed-typed language system supporting both typed and untyped modules with cross-module imports; enforced soundness using runtime proxies to catch type violations at module boundaries
- Architected CESK machine using Strategy pattern with 25 state transition cases, improving code maintainability by delegating each transition type to separate strategy classes
- Developed comprehensive test suite with JUnit unit tests and 50+ integration tests written in the target language; created Bash test runner script to automate test execution and failure reporting

Reversi Game (Java, Java Swing, JUnit) | Oct. 2023 - Dec. 2023

- Developed fully functional Reversi game (~2,000 lines) applying MVC architecture and design patterns including Observer, Decorator, and Strategy to ensure extensibility
 - Built GUI using Java Swing with custom components for board rendering, move validation, and game state visualization
 - Wrote 50+ unit tests using JUnit achieving 95%+ code coverage, validating game logic for edge cases including corner captures and board-full scenarios
-

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Skills

Languages: TypeScript, JavaScript, Java, Python, SQL, HTML, CSS

Frameworks & Tools: React, Next.js, Angular, Spring Boot, Node.js, Express.js, Docker, Git, GitHub Actions

Databases: MongoDB, MySQL, Redis

Development Practices: Agile/Scrum, CI/CD, Pair Programming, Test-Driven Development, RESTful API Design

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Claude resume formatted better:

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